

Air

MCQ

1. Air is a mixture of

- A) Only oxygen
- B) Only nitrogen
- C) Several gases
- D) Only carbon dioxide

Answer: C

2. The major component of air is

- A) Oxygen
- B) Nitrogen
- C) Carbon dioxide
- D) Hydrogen

Answer: B

3. Percentage of nitrogen in air is about

- A) 21%
- B) 50%
- C) 78%
- D) 90%

Answer: C

4. Percentage of oxygen in air is about

- A) 78%
- B) 21%
- C) 11%
- D) 1%

Answer: B

5. Air has no apparent

- A) Weight
- B) Pressure
- C) Colour and smell

D) Volume

Answer: C

6. Layers of air surrounding the earth are called

A) Biosphere

B) Lithosphere

C) Atmosphere

D) Hydrosphere

Answer: C

7. Plants need air mainly for

A) Photosynthesis and respiration

B) Drinking

C) Movement

D) Growth of roots only

Answer: A

8. Air is required for

A) Burning

B) Freezing

C) Melting

D) Condensation

Answer: A

9. Balloons are filled with

A) Water

B) Sand

C) Air

D) Oil

Answer: C

10. Birds fly because of

A) Gravity

B) Air

C) Water

D) Land

Answer: B

11. Oxygen was first discovered by

- A) Dalton
- B) C.W. Scheele
- C) Rutherford
- D) Lavoisier

Answer: B

12. Oxygen was named by

- A) Scheele
- B) Priestley
- C) Lavoisier
- D) Rutherford

Answer: C

13. Oxygen supports

- A) Rusting only
- B) Life and burning
- C) Only respiration
- D) Only photosynthesis

Answer: B

14. Oxygen is

- A) Combustible
- B) Non-combustible
- C) Explosive
- D) Poisonous

Answer: B

15. Oxygen actively supports

- A) Freezing
- B) Melting
- C) Combustion
- D) Condensation

Answer: C

16. Oxygen exists in free state as

- A) O

- B) O₂
- C) O₃
- D) H₂O

Answer: B

17. Ozone is represented as

- A) O
- B) O₂
- C) O₃
- D) CO₂

Answer: C

18. Ozone is present in

- A) Lower atmosphere
- B) Soil
- C) Upper atmosphere
- D) Water

Answer: C

19. Oxygen is most abundant in earth's crust by

- A) Volume
- B) Mass
- C) Number
- D) Density

Answer: B

20. Percentage of oxygen in atmosphere by volume is

- A) 78%
- B) 46%
- C) 21%
- D) 89%

Answer: C

21. Oxygen is _____ than air in density

- A) Lighter
- B) Equal
- C) Heavier

D) Much lighter

Answer: C

22. Oxygen is a poor conductor of

A) Sound

B) Heat and electricity

C) Light

D) Magnetism

Answer: B

23. Oxygen is slightly soluble in

A) Oil

B) Mercury

C) Water

D) Sand

Answer: C

24. Liquefaction of oxygen requires

A) High temperature

B) Low pressure

C) High pressure and low temperature

D) Normal conditions

Answer: C

25. Oxygen reacts with metals to form

A) Acids

B) Bases

C) Metal oxides

D) Salts

Answer: C

26. Metal oxides formed by oxygen are generally

A) Acidic

B) Basic

C) Neutral

D) Salty

Answer: B

27. Non-metals react with oxygen to form

- A) Basic oxides
- B) Acidic oxides
- C) Metal oxides
- D) Neutral salts

Answer: B

28. Burning of fuels is a reaction with

- A) Nitrogen
- B) Carbon dioxide
- C) Oxygen
- D) Hydrogen

Answer: C

29. Rusting occurs in presence of

- A) Air and moisture
- B) Only water
- C) Only air
- D) Acid

Answer: A

30. Rust is

- A) Ferric chloride
- B) Ferric sulphate
- C) Hydrated ferric oxide
- D) Iron carbonate

Answer: C

31. Oxygen is used in welding as

- A) Liquid oxygen
- B) Oxy-acetylene
- C) Compressed air
- D) Steam

Answer: B

32. Oxygen is used by living organisms for

- A) Nutrition

- B) Respiration
- C) Excretion
- D) Growth only

Answer: B

33. Nitrogen was discovered by

- A) Lavoisier
- B) Priestley
- C) Daniel Rutherford
- D) Dalton

Answer: C

34. Nitrogen is essential for

- A) Bones
- B) Proteins
- C) Water
- D) Fats

Answer: B

35. Nitrogen is present in nucleic acids like

- A) DNA and RNA
- B) Glucose
- C) Starch
- D) Cellulose

Answer: A

36. Nitrogen exists freely in air as

- A) N
- B) NO₂
- C) N₂
- D) NH₃

Answer: C

37. Nitrogen makes up about _____ of air

- A) 21%
- B) 78%
- C) 50%

D) 1%

Answer: B

38. Nitrogen is _____ to litmus

A) Acidic

B) Basic

C) Neutral

D) Salty

Answer: C

39. Nitrogen is slightly _____ than air

A) Heavier

B) Lighter

C) Equal

D) Denser

Answer: B

40. Liquid nitrogen is used as

A) Fuel

B) Refrigerant

C) Fertilizer

D) Lubricant

Answer: B

41. Nitrogen does not support

A) Respiration

B) Photosynthesis

C) Combustion

D) Growth

Answer: C

42. Nitrogen reacts with hydrogen to form

A) Nitric acid

B) Ammonia

C) Nitrogen dioxide

D) Urea

Answer: B

43. Haber process is used to prepare

- A) Nitric acid
- B) Ammonia
- C) Urea
- D) Nitrates

Answer: B

44. Nitrogen fixation converts nitrogen into

- A) Inert form
- B) Useful compounds
- C) Metals
- D) Salts only

Answer: B

45. Nitrogen fixing bacteria live in

- A) Leaves
- B) Flowers
- C) Root nodules
- D) Stem

Answer: C

46. Carbon dioxide contains

- A) One carbon and one oxygen
- B) One carbon and two oxygen
- C) Two carbon and one oxygen
- D) Three oxygen

Answer: B

47. Percentage of CO₂ in air is about

- A) 21%
- B) 78%
- C) 0.03%
- D) 1%

Answer: C

48. CO₂ is produced during

- A) Respiration

B) Photosynthesis

C) Freezing

D) Condensation

Answer: A

49. Carbon dioxide is _____ than air

A) Lighter

B) Equal

C) Heavier

D) Much lighter

Answer: C

50. Carbon dioxide does not support

A) Respiration

B) Combustion

C) Photosynthesis

D) Fermentation

Answer: B

51. CO₂ turns blue litmus

A) Blue

B) Red

C) Green

D) Yellow

Answer: B

52. Solid CO₂ is called

A) Ice

B) Snow

C) Dry ice

D) Frost

Answer: C

53. Dry ice undergoes

A) Melting

B) Evaporation

C) Sublimation

D) Condensation

Answer: C

54. CO₂ is used in

A) Fire extinguishers

B) Cooking

C) Welding

D) Lighting

Answer: A

55. Lime water turns milky due to formation of

A) Calcium hydroxide

B) Calcium carbonate

C) Calcium oxide

D) Calcium chloride

Answer: B

56. Excess CO₂ turns milky clear due to formation of

A) CaCO₃

B) CaO

C) Ca(HCO₃)₂

D) CaCl₂

Answer: C

57. Greenhouse effect traps

A) Cold air

B) Sun's heat

C) Oxygen

D) Nitrogen

Answer: B

58. Major greenhouse gas is

A) Nitrogen

B) Oxygen

C) Carbon dioxide

D) Hydrogen

Answer: C

59. Global warming is due to increase in

- A) Oxygen
- B) Nitrogen
- C) Greenhouse gases
- D) Rainfall

Answer: C

60. One effect of global warming is

- A) Ice formation
- B) Melting of glaciers
- C) Decrease in sea level
- D) Less rainfall always

Answer: B

61. Global warming causes rise in

- A) Soil fertility
- B) Sea level
- C) Oxygen level
- D) Nitrogen level

Answer: B

62. Deforestation leads to

- A) Less CO₂
- B) More CO₂
- C) More oxygen
- D) No change

Answer: B

63. Planting trees helps to reduce

- A) Oxygen
- B) Nitrogen
- C) Carbon dioxide
- D) Water vapour

Answer: C

64. Acid rain is caused by

- A) CO₂ only

B) SO₂ and NO₂

C) Oxygen

D) Nitrogen

Answer: B

65. pH of acid rain is

A) More than 7

B) Equal to 7

C) Less than 5.6

D) Equal to 5.6

Answer: C

66. Acid rain damages

A) Only animals

B) Only plants

C) Buildings and plants

D) Air only

Answer: C

67. Acid rain affects soil by

A) Increasing fertility

B) No change

C) Decreasing fertility

D) Making it neutral

Answer: C

68. Use of CNG helps reduce

A) Acid rain

B) Oxygen

C) Nitrogen

D) Water

Answer: A

69. Using renewable energy helps reduce

A) Pollution

B) Rainfall

C) Wind

D) Sunlight

Answer: A

70. One preventive measure of global warming is

A) Deforestation

B) Using fossil fuels

C) Recycling

D) Burning plastics

Answer: C

71. Acid rain irritates

A) Hair

B) Eyes and skin

C) Bones

D) Muscles

Answer: B

72. Nitrogen in air helps by

A) Supporting burning

B) Moderating combustion

C) Increasing fire

D) Causing rust

Answer: B

73. Nitrogen is chemically _____ under normal conditions

A) Active

B) Reactive

C) Inert

D) Explosive

Answer: C

74. CO₂ is used to prepare

A) Soap

B) Soft drinks

C) Cement

D) Plastics

Answer: B

75. Solvay process produces

- A) Sodium chloride
- B) Sodium carbonate
- C) Sodium bicarbonate
- D) Calcium oxide

Answer: B

76. CO₂ is used with ammonia to make

- A) Urea
- B) Soap
- C) Paint
- D) Glass

Answer: A

77. Acid rain corrodes

- A) Plastic
- B) Rubber
- C) Buildings
- D) Wood only

Answer: C

78. Nitrogen in tyres helps

- A) Increase pressure quickly
- B) Reduce wear and tear
- C) Burn fuel
- D) Increase speed

Answer: B

79. Oxygen is used in synthesis of

- A) Sugar
- B) Methanol
- C) Salt
- D) Water

Answer: B

80. Photosynthesis releases

- A) Carbon dioxide

- B) Nitrogen
- C) Oxygen
- D) Hydrogen

Answer: C

81. CO₂ is acidic because it forms

- A) Base in water
- B) Acid in water
- C) Salt in water
- D) Neutral solution

Answer: B

82. Oxygen is essential for

- A) Combustion only
- B) Respiration only
- C) Both respiration and combustion
- D) Neither

Answer: C

83. Nitrogen compounds are used mainly as

- A) Fuels
- B) Fertilizers
- C) Lubricants
- D) Dyes

Answer: B

84. Liquid nitrogen appears

- A) Yellow
- B) Blue
- C) Colourless like water
- D) Green

Answer: C

85. CO₂ fire extinguishers work by

- A) Cooling only
- B) Removing oxygen
- C) Adding oxygen

D) Adding fuel

Answer: B

86. Ozone protects life by absorbing

A) Infrared rays

B) Visible light

C) Ultraviolet rays

D) X-rays

Answer: C

87. Nitrogen in food is required for

A) Energy

B) Protein formation

C) Taste

D) Colour

Answer: B

88. Acid rain is prevented by reducing

A) Oxygen

B) Sulphur dioxide

C) Nitrogen gas

D) Water vapour

Answer: B

89. CO₂ preservation of food works by

A) Increasing oxygen

B) Preventing microbes

C) Adding sweetness

D) Drying food

Answer: B

90. Greenhouse gases trap heat in

A) Lithosphere

B) Atmosphere

C) Hydrosphere

D) Biosphere

Answer: B

91. Nitrogen reacts with metals at

- A) Low temperature
- B) Room temperature
- C) High temperature
- D) Freezing point

Answer: C

92. Nitrogen dioxide is formed from

- A) Nitrogen and hydrogen
- B) Nitrogen and oxygen
- C) Nitrogen and carbon
- D) Nitrogen and sulphur

Answer: B

93. Acid rain affects aquatic life by

- A) Increasing oxygen
- B) Lowering pH
- C) Increasing temperature
- D) Increasing nitrogen

Answer: B

94. Oxygen is slightly soluble in

- A) Oil
- B) Alcohol
- C) Water
- D) Mercury

Answer: C

95. Nitrogen is used in bulbs to

- A) Increase light
- B) Reduce evaporation of filament
- C) Increase heat
- D) Produce colour

Answer: B

96. CO₂ fog is used in

- A) Laboratories

- B) Hospitals
- C) Stage shows
- D) Schools

Answer: C

97. Nitrogen is present in human body about

- A) 1%
- B) 3%
- C) 10%
- D) 21%

Answer: B

98. Greenhouse effect is necessary to

- A) Destroy life
- B) Freeze earth
- C) Maintain earth's temperature
- D) Stop rainfall

Answer: C

99. Acid rain mainly affects

- A) Oceans only
- B) Forests and monuments
- C) Deserts
- D) Volcanoes

Answer: B

100. Air pollution increases

- A) Oxygen level
- B) Greenhouse gases
- C) Nitrogen fixation
- D) Soil fertility

Answer: B